**In this question you must show all stages of your working.**

**Solutions relying entirely on calculator technology are not acceptable.**

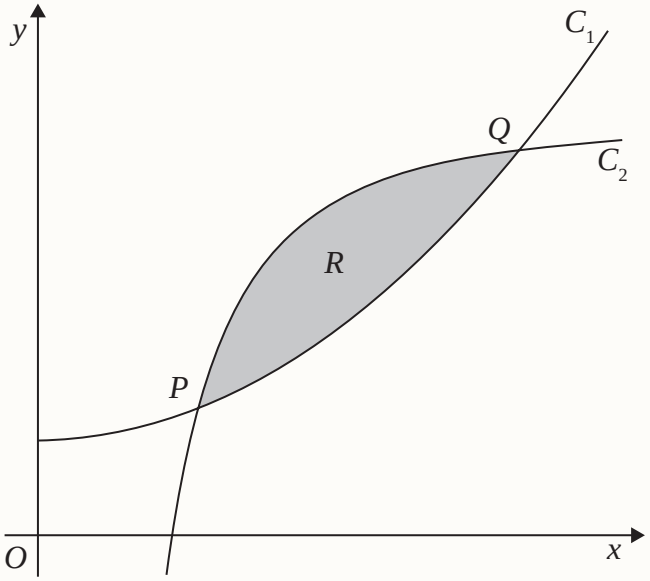



**Figure 1**

Figure 1 shows a sketch of part of the curve $C_1$ with equation

$$y = x^2 + 3 \qquad x > 0$$

and part of the curve $C_2$ with equation

$$y = 13 - \frac{9}{x^2} \qquad x > 0$$

The curves $C_1$ and $C_2$ intersect at the points $P$ and $Q$ as shown in Figure 1.

(a) Use algebra to find the $x$ coordinate of $P$ and the $x$ coordinate of $Q$.

**(4)**

The finite region $R$, shown shaded in Figure 1, is bounded by $C_1$ and $C_2$

(b) Use algebraic integration to find the exact area of $R$.

**(4)**